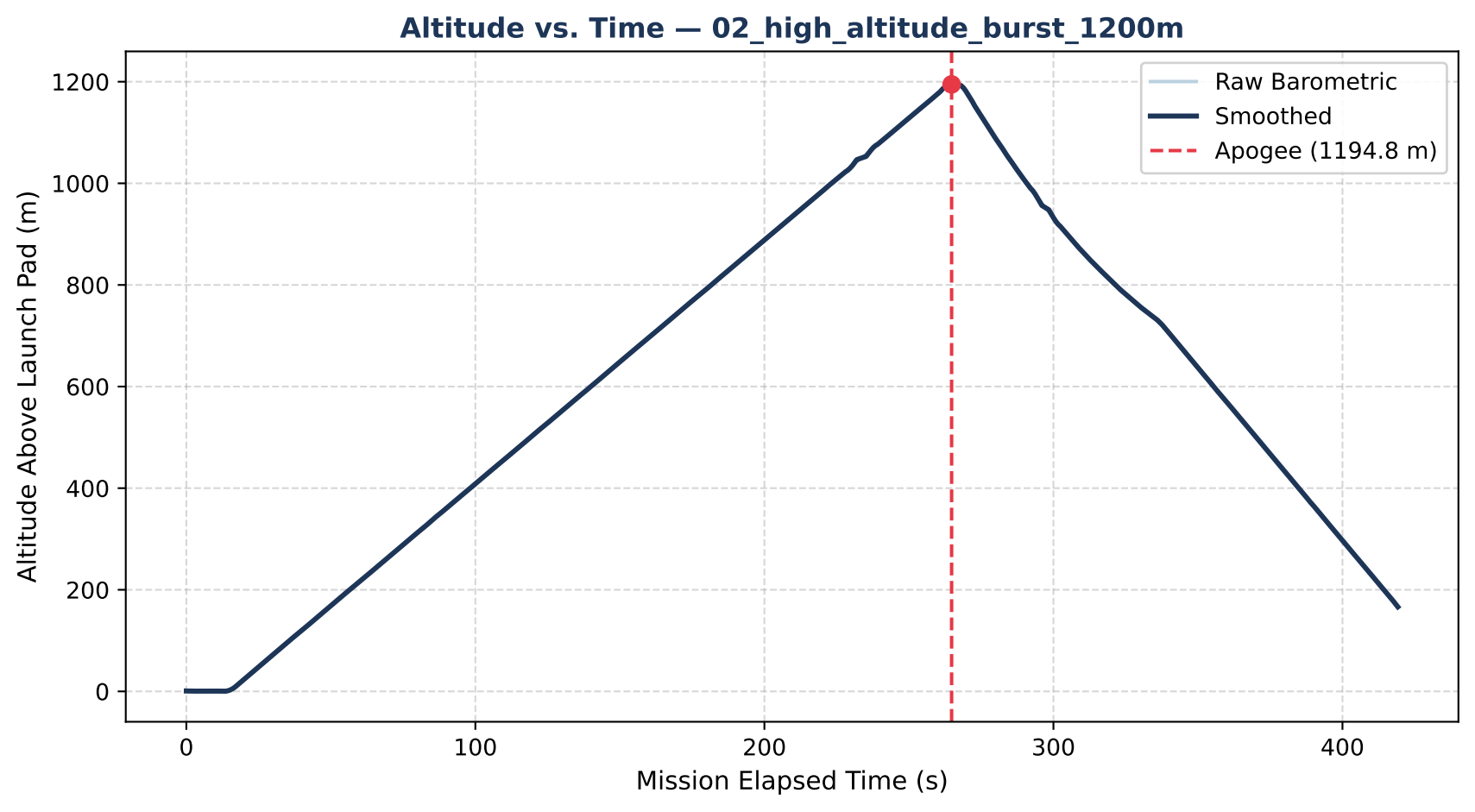
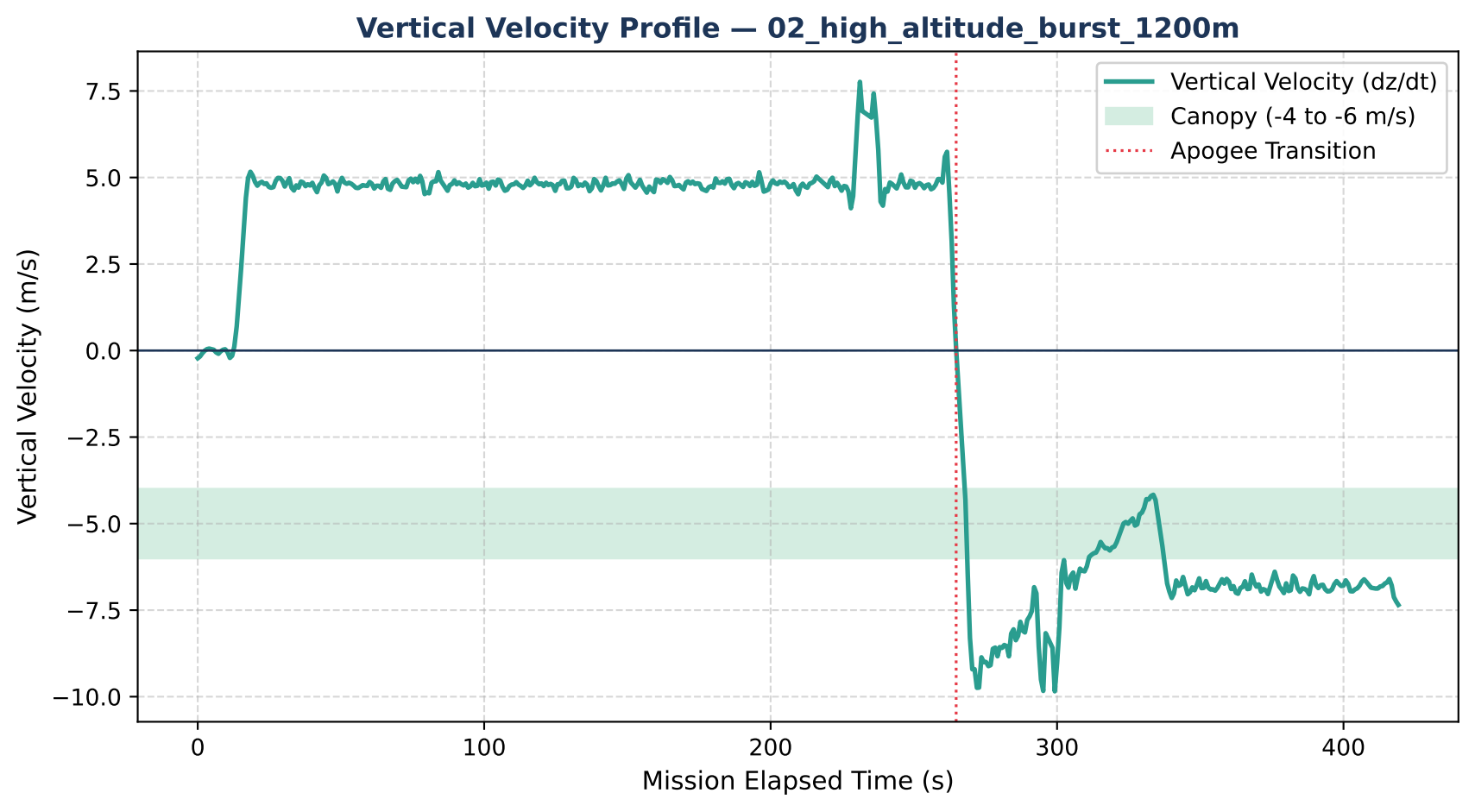
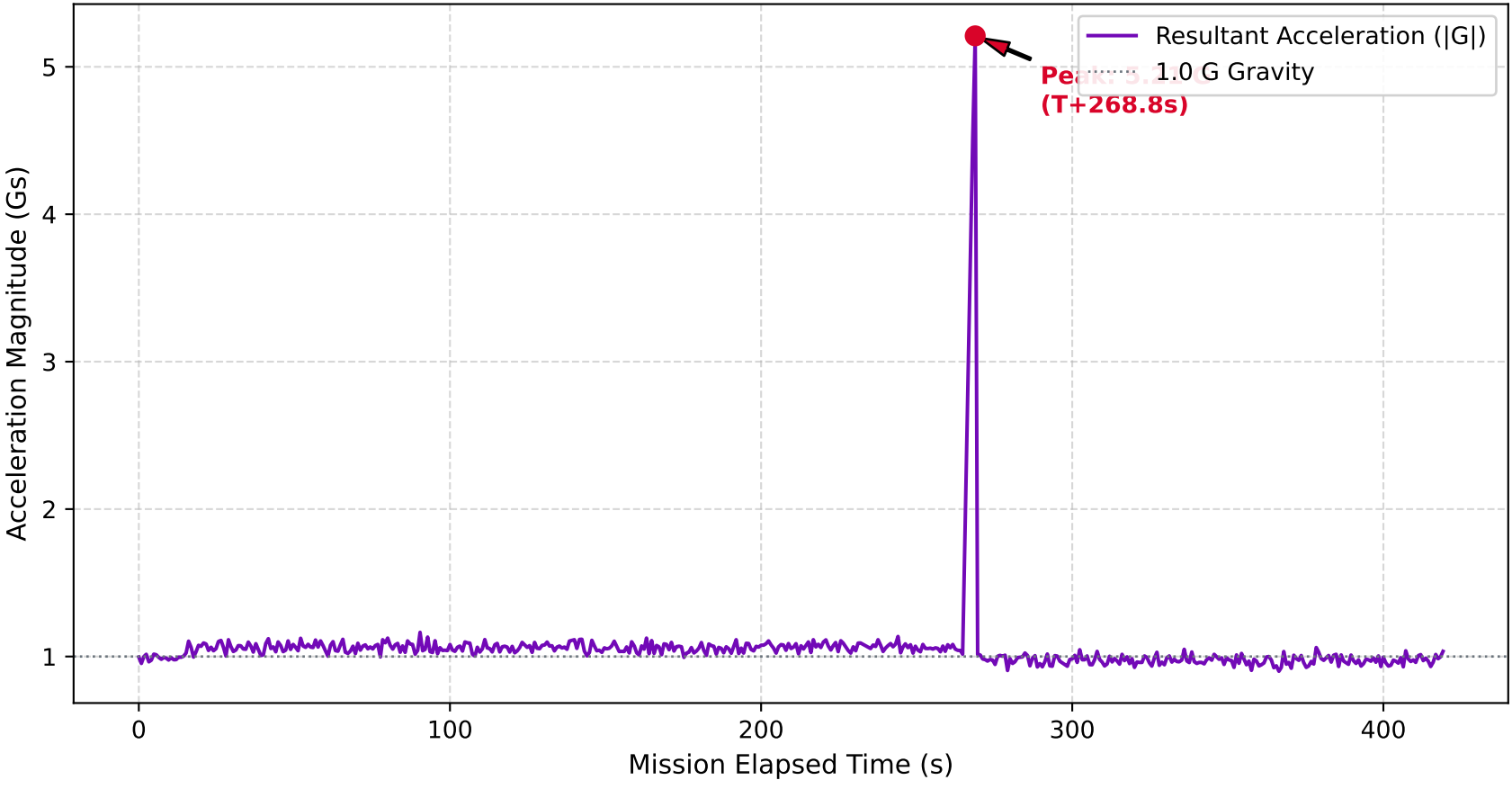
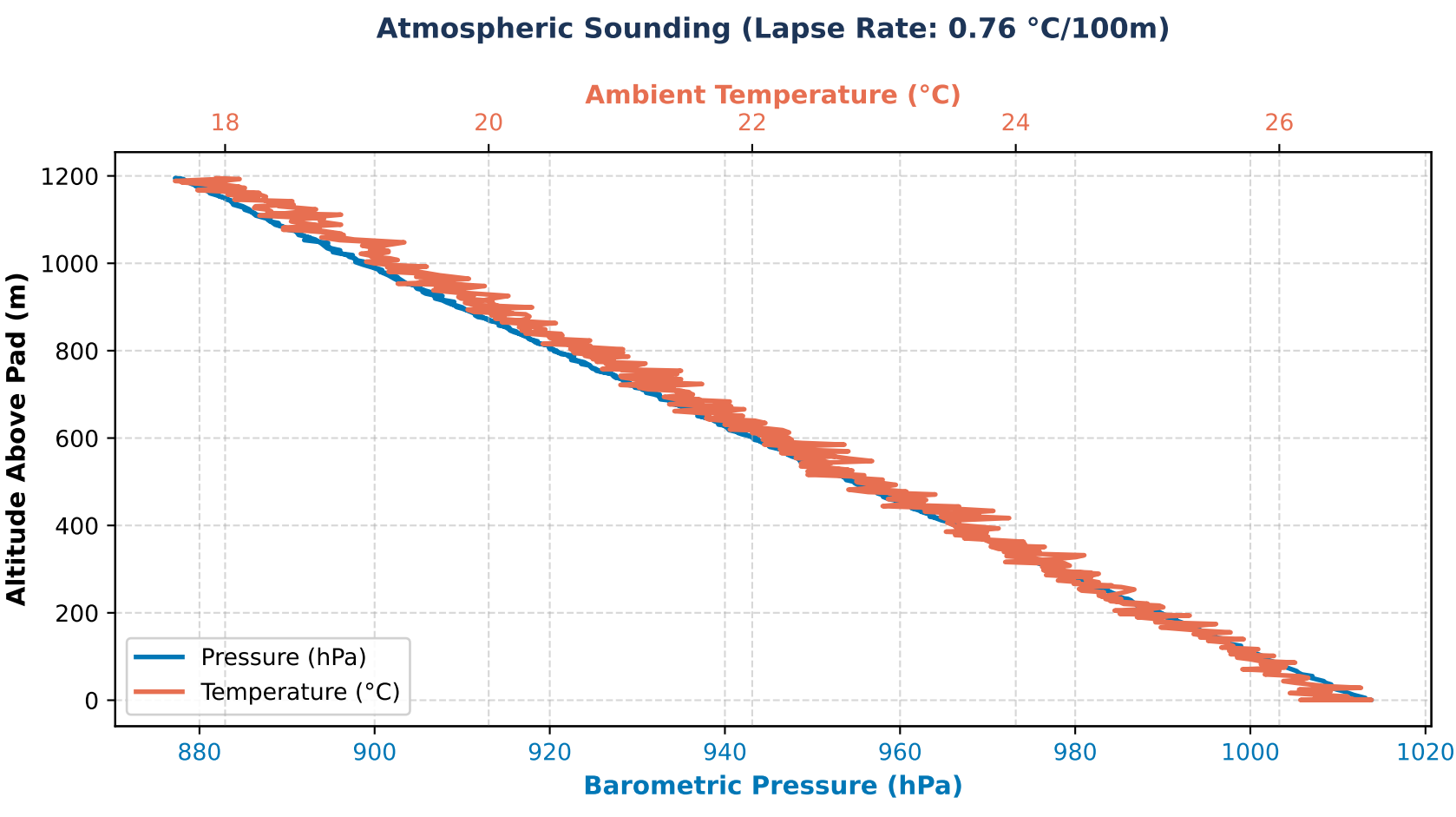
Cognitive CanSat - Mission Summary: 02_high_altitude_burst_1200m

Flight Dynamics Parameter	Calibrated Value
Mission Name	02_high_altitude_burst_1200m
Apogee (m)	1194.75
Time to Apogee (s)	264.77
Max Ascent Velocity (m/s)	7.76
Max Descent Velocity (m/s)	-9.84
Max G-Shock (Gs)	5.21
Touchdown Shock (Gs)	1.06
Mission Duration (s)	419.16
Min Temperature (°C)	17.62
Max Temperature (°C)	26.7
Min Pressure (hPa)	877.21
Max Pressure (hPa)	1013.84
Lapse Rate (°C/100m)	0.76
Telemetry Frames	517
Landing Lat	22.578513
Landing Lon	88.372014


Altitude vs. Time — 02_high_altitude_burst_1200m
Raw Barometric
Smoothed
Apogee (1194.8 m)
Altitude Above Launch Pad (m)
Mission Elapsed Time (s)
0
200
400
600
800
1000
1200
0
100
200
300
400


Vertical Velocity Profile — 02_high_altitude_burst_1200m
Vertical Velocity (dz/dt)
Canopy (-4 to -6 m/s)
Apogee Transition
Vertical Velocity (m/s)
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
0
100
200
300
400
Mission Elapsed Time (s)


G-Shock & Acceleration — 02_high_altitude_burst_1200m
Resultant Acceleration (|G|)
1.0 G Gravity
(T+268.8s)
Acceleration Magnitude (Gs)
Mission Elapsed Time (s)
1
2
3
4
5
0
100
200
300
400


Atmospheric Sounding (Lapse Rate: 0.76 °C/100m)
Ambient Temperature (°C)
18
20
22
24
26
Altitude Above Pad (m)
1200
1000
800
600
400
200
0
Pressure (hPa)
Temperature (°C)
880
900
920
940
960
980
1000
1020
Barometric Pressure (hPa)

Cognitive CanSat Mission Dynamics: 02_high_altitude_burst_1200m

